

TASIDO F2 ZONAL ASSESSMENT 2026 SOLUTIONS

1. Multiple choice questions

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
D	B	C	B	D	A	A	B	C	Free

(x) Capacitance in series is given by

$$1/C = 1/C_1 + 1/C_2 + 1/C_3$$

$$\frac{1}{C} = \frac{1}{1\mu F} + \frac{1}{4\mu F} + \frac{1}{2\mu F}$$

$$\frac{1}{C} = \frac{4+1+2}{4} \times \frac{1}{\mu F}$$

$$\frac{1}{C} = \frac{7}{4 \mu F}$$

$$C = \frac{4 \mu F}{7}$$

2. Matching item question

List A	(i)	(ii)	(iii)	(iv)	(v)
List B	B	E	D	H	G

3. (a) Archimedes' principle states that "When a body is totally or partially immersed in a fluid, it experiences an upthrust which is equal to the weight of the fluid displaced by the body".

(b) (i) Real weight =?

From

$$\text{Upthrust} = \text{Real weight} - \text{Apparent weight}$$

$$2 \text{ N} = \text{Real weight} - 18 \text{ N}$$

$$\text{Real weight} = 2 \text{ N} + 18 \text{ N} = 20 \text{ N}$$

The real weight of an object is 20 N.

(ii) Relative density of an object =?

From

$$\text{R.D of an object} = \frac{\text{Real weight}}{\text{Upthrust}}$$

$$\text{R.D of an object} = \frac{20 \text{ N}}{2 \text{ N}} = 10$$

Therefore, the relative density of an object is 10.

(iii) Density of an object in kg/m³

From

$$\text{R.D of an object} = \frac{\text{Density of an object}}{\text{Density of water}}$$

$$10 \times \text{density of water} = \text{density of an object}$$

$$10 \times 1000 \text{ kg/m}^3 = 10000 \text{ kg/m}^3 = \text{Density of an object}$$

Therefore, the density of an object is 10000 kg/m³.

4. (a) Density of a substance is the mass of a substance per unit its volume while relative density is the ratio of density of substance to the density of water. Density is measured in kg/m³ while relative density has no unit.

(b) Mass of empty density bottle, $m_0 = 12.0 \text{ g}$

Mass of bottle containing sand, $m_1 = 62.0 \text{ g}$

Mass of bottle containing sand and water, $m_2 = 132.0 \text{ g}$

Mass of bottle filled with water, $m_3 = 87.0 \text{ g}$

(i) Relative density of sand =?

From

$$\text{Relative density of sand} = (m_1 - m_0) / ((m_3 - m_0) - (m_2 - m_1))$$

$$\text{Relative density of sand} = \frac{62.0 \text{ g} - 12.0 \text{ g}}{(87.0 \text{ g} - 12.0 \text{ g}) - (132.0 \text{ g} - 62.0 \text{ g})}$$

$$\text{Relative density of sand} = \frac{50.0 \text{ g}}{5.0 \text{ g}} = 10$$

The relative density of sand is 10.

(ii) Density of sand =?

From

$$\text{R.D of sand} = \frac{\text{Density of sand}}{\text{Density of water}}$$

$$10 \times \text{density of water} = \text{density sand}$$

$$10 \times 1000 \text{ kg/m}^3 = 10000 \text{ kg/m}^3 = \text{Density of sand}$$

Therefore, the density of sand is 10000 kg/m³.

(iii) Volume of sand =?

$$\text{Volume of sand} = \frac{\text{Mass of the sand}}{\text{Density of the sand}}$$

$$\text{Mass of sand} = m_1 - m_0 = 62.0 \text{ g} - 12.0 \text{ g} = 50 \text{ g} = 0.05 \text{ kg}$$

$$\text{Volume of the sand} = 0.05 \text{ kg}/10000 \text{ kgm}^{-3}$$

$$\text{Volume of the sand} = 5.0 \times 10^{-6} \text{ m}^3$$

Therefore, the volume of the sand is $5.0 \times 10^{-6} \text{ m}^3$.

(iv) Volume of the bottle =?

$$\text{Volume of bottle} = \text{volume of water}$$

$$\text{Volume of water} = \frac{\text{Mass of water}}{\text{Density of water}}$$

$$\text{Mass of water} = m_3 - m_0 = 87.0 \text{ g} - 12.0 \text{ g} = 75 \text{ g} = 0.075 \text{ kg}$$

$$\text{Volume of water} = 0.075 \text{ kg}/1000 \text{ kgm}^{-3}$$

$$\text{Volume of bottle} = \text{volume of water} = 7.5 \times 10^{-5} \text{ m}^3$$

5. (a) This is because a thin wire exerts high pressure on the soap due to its small area. This makes easier to cut a bar of soap.

(b) Pressure exerted by the car =?

From the formula

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$\text{Force} = \text{weight} = mg = 8000 \text{ kg} \times 10 \text{ N/kg} = 80000 \text{ N}$$

$$\text{Area} = 4 \times 50 \text{ cm}^2 = 0.02 \text{ m}^2$$

$$\text{Pressure} = 80000 \text{ N}/0.02 \text{ m}^2 = 4 \times 10^6 \text{ N/m}^2$$

Therefore, the pressure exerted by the car on the ground is $4 \times 10^6 \text{ N/m}^2$.

6. (a) (i) Energy possessed by a moving car and energy possessed by a moving water in a river.

(ii) Mass of a body and velocity of the body are the factors that determine the kinetic energy of a body.

(b) Mass, $m = 20 \text{ kg}$

Velocity, $v = 6 \text{ m/s}$

Time, $t = 10 \text{ s}$

(i) Power in watts

$$\text{Power} = \frac{\text{Work done}}{\text{time}} = \frac{\text{Kinetic energy}}{\text{time}}$$

$$\text{Power} = mv^2/2t = (20 \text{ kg} \times (6 \text{ m/s})^2)/(2 \times 10 \text{ s})$$

$$\text{Power} = 36 \text{ watts}$$

The power developed is 36 watts

(ii) Power in horsepower

$$1 \text{ hp} = 746 \text{ W}$$

$$P = 36 \text{ W}$$

$$P = \frac{36 \text{ hp}}{746} = 0.048 \text{ hp}$$

Therefore, the power developed was 0.048 hp.

7. (a) (i) The water meniscus bends downward because the adhesive force (force of attraction between water molecules and glass molecules) is stronger than cohesive force (force of attraction between water molecules themselves).

(ii) Meniscus of mercury bend upward because the cohesive force (force of attraction between mercury molecules) is stronger than adhesive force (force of attraction between mercury molecules and glass molecules).

(b) Data given

$$F_1 = 4 \text{ N}$$

$$e_1 = 1.0 \text{ mm}$$

$$F_2 = ? \text{ If } e_2 = 3.2 \text{ mm}$$

From Hooke's law

$$F = ke$$

$$F_1 = ke_1$$

$$F_2 = ke_2$$

$$F_2/F_1 = ke_2/ke_1$$

$$F_2 = F_1 \times (e_2/e_1) = 4.0 \text{ N} \times \frac{3.2 \text{ mm}}{1.0 \text{ mm}}$$

$$F_2 = 12.8 \text{ N}$$

Therefore, a force of 12.8 N will cause an extension of 3.2 mm.

8. (a) (i) Speed is the rate of distance covered by an object while velocity is the rate of distance covered in a specific direction.
- (ii) Speed is a scalar quantity while velocity is a vector quantity.
- (b) (i) Consider a velocity-time graph below

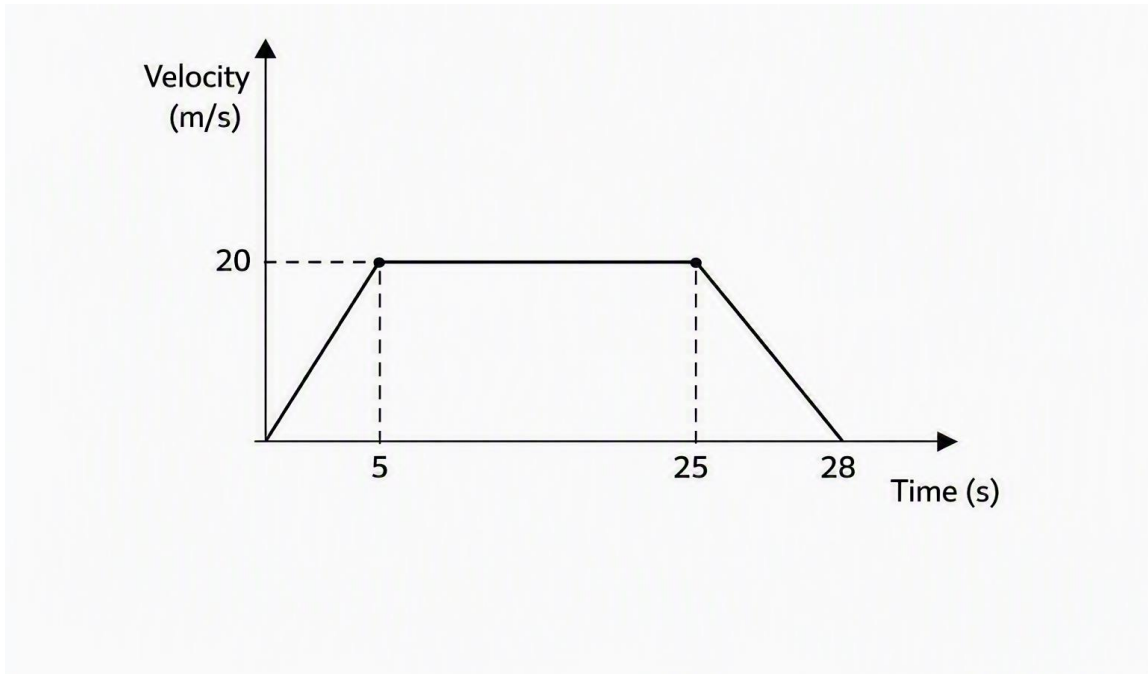


Fig. velocity-time graph

(ii) Total distance covered

From the velocity-time graph above

$$\text{Total distance covered} = \text{Area under the graph} = \frac{(a+b) \times h}{2}$$

Where, $a = 25 \text{ s} - 5 \text{ s} = 20 \text{ s}$, $b = 28 \text{ s}$ and $h = 20 \text{ m/s}$

$$\text{Total distance covered} = \frac{(20 \text{ s} + 28 \text{ s}) \times 20 \text{ m/s}}{2} = 480 \text{ m}$$

Therefore, the total distance covered was 480 m.

9. (a) Uses of magnets

- (i) Electric generators: These use permanent magnets to convert mechanical energy to electrical energy.
- (ii) Speakers and microphones: These use permanent magnets and current carrying coils to convert electric energy into sound and sound energy into electrical energy, respectively.
- (iii) Credit, debit and ATM cards: These have a magnetic strip on one of their sides. This strip contains the necessary information to contact an individual's financial institution and connect with their account.
- (iv) CRT (Cathode Ray Tube) television and computer monitors: These employ an electromagnet in their inputs to assist in producing an image on the screen

(b) magnets are installed in the doors of fridge and microwave so as to make the doors tighter and hence prevent the heat exchange with the surrounding.

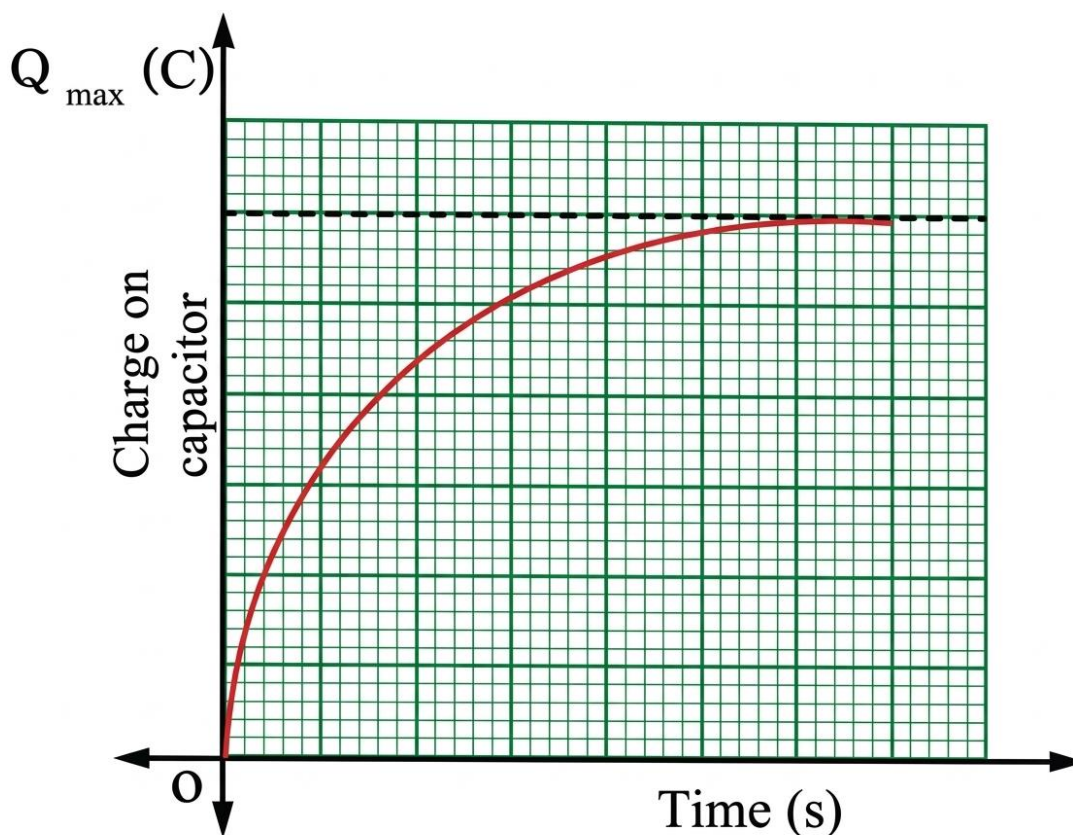
(c) Properties of magnets

- (i) All magnets have two poles, the north and south poles.
- (ii) Magnets exert a force on some materials and do not on others.
- (iii) The magnetic force is an action-at-a-distance force.
- (iv) When two magnets are brought together, like poles repel each other while unlike poles attract.

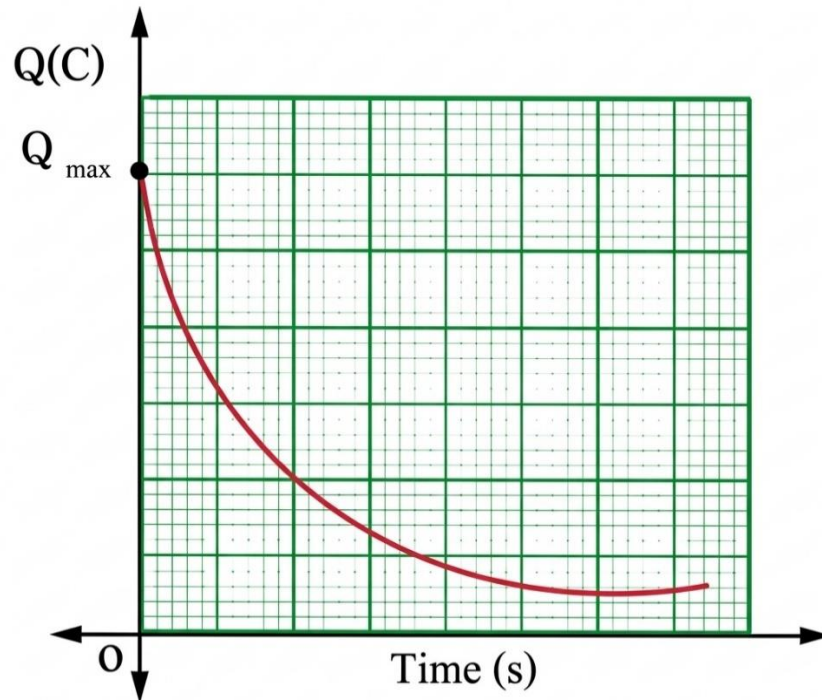
(d) Like poles repel while unlike poles attract each other. This means that to test polarity, bring a known North pole near the unknown magnet. If it repels, that end is also a North pole.

Repulsion is a sure test because magnetic materials such as iron can also be attracted when placed close to the magnet.

10. (a) (i) Charging a capacitor



(ii) Discharging a capacitor



(b) Factors affecting capacitance

(i) Area of the plates

(ii) Distance between the plates

(iii) Insulating property of a medium

(c) (i) Total capacitance, $C = ?$

From

$$C = C_1 + C_2$$

Where, $C_1 = 3 \mu\text{F}$ and $C_2 = 5 \mu\text{F}$

$$C = 3 \mu\text{F} + 5 \mu\text{F} = 8 \mu\text{F}$$

Therefore, the total capacitance is $8 \mu\text{F}$

(ii) Quantity of charge stored = ?

From

$$Q = CV$$

$$Q = 8 \mu\text{F} \times 30 \text{ V} = 240 \mu\text{C} = 0.24 \text{ mC}$$

Therefore, the amount of charge stored was 0.24 mC